



Week 2 Lecture 4

▼ Class	BSCCS2003
🕒 Created	@September 14, 2021 7:21 PM
🔗 Materials	
# Module #	13
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Introduction to HTML

Topics

- HyperText Markup Language
- Generalizations
- Variants of Interest

HyperText Markup Language (HTML)

- HTML was first used by Tim Berners-Lee in the original Web at CERN (-1989)
- Considered an application of SGML (Standard Generalized Markup Language)
 - Strict definitions on structure, syntax and validity
- HTML was meant for browser interpretation
 - Very forgiving: loose validity checks
 - Best effort to display

HTML Example

```
<!DOCTYPE html>
<html>
<body>
<h1>My First Heading.</h1>
<p>My first paragraph.</p>
</body>
</html>
```

Tags

- `<h1></h1>` - paired tags
- Represented by angled brackets `<>`
- Closing tag with `/`
- Location specific: `<DOCTYPE>` is only at the head of the document
- Case insensitive

Nesting

Valid syntax

- `<em><strong>Hello</strong></em>`

- The output is

- ***Hello***

- Invalid:

- `<em><strong>Hello</em></strong>`

- `<em><strong>Hello</em>`

- `<em><strong>Hell<o/em></strong>`

Presentation v/s Semantics

- `<strong>Hello</strong>`

- `<b>Hello</b>`

- **Hello**

Which one is right? Which one is better?

It is not an easy question to answer, both of them may be correct

`<strong>` is putting an emphasis meanwhile `<b>` does not

Timelines

- SGML based
 - 1989 - HTML original
 - 1995 - HTML 2
 - 1997 - HTML 3, 4
- XML based
 - XHTML - 1997 - mid 2010s
- HTML 5
 - first release in 2008
 - W3C recommendation - 2014
 - World Wide Web Consortium

HTML 5

- Block element: `<div>`
- In-line element: `<span>`
- Logical elements: `<nav>`, `<footer>`
- Media: `<audio>`, `<video>`

Remove "presentation only" tags:

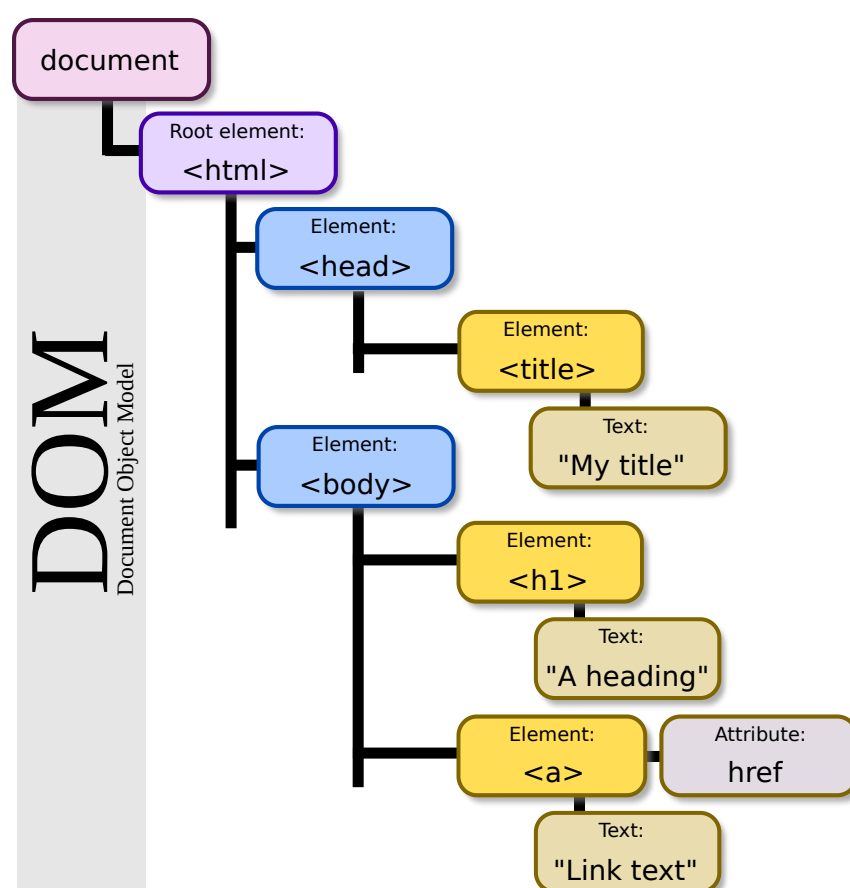
- `<center>`
- `<font>`

Document Object Model (DOM)

The Document Object Model (DOM) is a programming interface for web documents. It represents the page so that programs can change the document structure, style, and content. The DOM represents the document as nodes and objects; that way, programming languages can interact with the page.

Source: https://developer.mozilla.org/en-US/docs/Web/API/Document_Object_Model/Introduction

```
<html>
<head>
  <title>My title</title>
</head>
<body>
  <h1>A heading</h1>
  <a href="https://onlinedegree.iitm.ac.in">Link text</a>
</body>
</html>
```



Source: https://en.wikipedia.org/wiki/Document_Object_Model

- It is a tree structure which represents the logical layout of the document
- Direct manipulation of this tree is possible!
- Application Programming Interfaces (APIs) are also defined on the DOM:
 - Canvas
 - Offline
 - Web Storage
 - Drag & Drop
 - ...
- In most of the cases, when we talk about manipulating the DOM, we use JavaScript
- Cascading Style Sheets (CSS) is used for styling